Linux Processes

Day 1 Investigation Worksheet — detailed responses

Linux process

What did I ask?	What is a Linux process?
What did AI tell me?	A Linux process is an instance of a running program. Each process generally has a unique process ID (PID) and uses resources such as CPU time and memory while executing.
What did I expect?	A running program managed by the operating system and identifiable by its process ID.
What command/action did I perform?	ps; sleep 60 &; ps
What happened?	Inspected the current processes, started a background process using sleep 60 &, and checked the process listing to observe the running command and its PID.
Yes / No / Partially (AI correctness)	Yes
What did I learn?	Processes are how the operating system manages running programs. Each process has an identity and state, and process inspection helps developers understand what is running.

ps

What did I ask?	What information does the ps command display in Linux?
What did AI tell me?	ps displays a snapshot of process information. Without additional options, it generally shows processes associated with the current terminal, including their PIDs, terminal information, CPU time, and command names.
What did I expect?	A command that lists currently running processes and provides basic information about them.
What command/action did I perform?	ps
What happened?	The terminal displayed a snapshot of processes associated with the current terminal session, including their process IDs and command names.
Yes / No / Partially (AI correctness)	Yes
What did I learn?	ps is useful for checking processes in the current terminal context. Additional options are needed when investigating processes across the entire system.

ps aux

What did I ask?	What information does the ps aux command display in Linux?
What did AI tell me?	ps aux displays a detailed snapshot of processes across the system. It includes the process owner, PID, CPU and memory usage, virtual and resident memory, process state, start information, CPU time, and command.
What did I expect?	A detailed listing of processes from all users, along with resource usage and identifying information.
What command/action did I perform?	ps aux

What happened?	The terminal displayed a system-wide process listing with details such as users, PIDs, CPU and memory usage, process states, and commands.
Yes / No / Partially (AI correctness)	Yes
What did I learn?	ps aux helps investigate processes across the system and identify applications that may be consuming excessive resources. Its output is a snapshot rather than a continuously updating display.

top

What did I ask?	What is the purpose of the top command in Linux?
What did AI tell me?	top provides an interactive, continuously updating view of running processes and system resource usage. It displays CPU utilization, memory usage, process IDs, process states, and resource consumption.
What did I expect?	A tool that displays live process information and helps monitor CPU and memory usage.
What command/action did I perform?	top
What happened?	The terminal opened an interactive display showing system resource statistics and a list of processes. The information updated periodically while the command was running.
Yes / No / Partially (AI correctness)	Yes
What did I learn?	top is useful for monitoring system performance and spotting processes that consume unusually high CPU or memory. Pressing q exits the interactive display.

Why investigate processes?

What did I ask?	Why would a developer investigate running processes in Linux?
What did AI tell me?	Developers inspect processes to troubleshoot performance problems, identify resource-intensive applications, investigate unresponsive services, verify that applications are running, and diagnose unexpected behavior.
What did I expect?	Process inspection to be a troubleshooting technique for identifying application problems and understanding resource consumption.
What command/action did I perform?	ps aux; top
What happened?	Used process-listing and monitoring commands to inspect running applications and view their CPU and memory consumption. These commands can help narrow down performance issues.
Yes / No / Partially (AI correctness)	Yes
What did I learn?	Process investigation helps identify potential causes of system slowdowns, high resource usage, and service failures. Process information should be combined with application logs and other diagnostic tools to determine the root cause.

Challenge: Slow application

What did I ask?	If an application is running slowly, can the top command help investigate the problem?

What did AI tell me?	Yes. top can show whether a process is consuming excessive CPU or memory and whether the system is under resource pressure. It provides a live view of process activity and overall system utilization.
What did I expect?	A tool that helps identify resource bottlenecks and provides evidence about possible causes of an application's poor performance.
What command/action did I perform?	top; ps aux
What happened?	Inspected system-wide resource usage and process-level CPU and memory consumption. This information can help identify resource-intensive processes and determine whether CPU or memory pressure might contribute to the slowdown.
Yes / No / Partially (AI correctness)	Yes
What did I learn?	top helps investigate performance problems by showing CPU usage, memory usage, process states, and system load. It cannot always identify the root cause; application logs, network diagnostics, disk activity tools, or profilers may also be needed.

Final Learning

The investigation helped me understand how Linux manages running programs and how ps and top can help identify resource usage and troubleshoot performance problems. Process monitoring is a starting point; logs and additional diagnostics may be needed to find the root cause.